Computer Vision

Pre-Processing

Technische Hochschule Rosenheim
Winter 2026/27
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Overview

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- Histograms and threshold values (binarization)
- linear filters
  - High and low pass filter
- non-linear filters
  - Ranking operations
  - Morphological operations
- Standardization
  - Size
  - Moments
  - Brightness

Pre-processing – Objective

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Transformation of a pattern so that

- further processing of the pattern becomes **easier**
- or the result is **better**
- or both

In pre-processing, a pattern is transformed into a new pattern – **an image is transformed to a new image**

Conversion to Grayscale

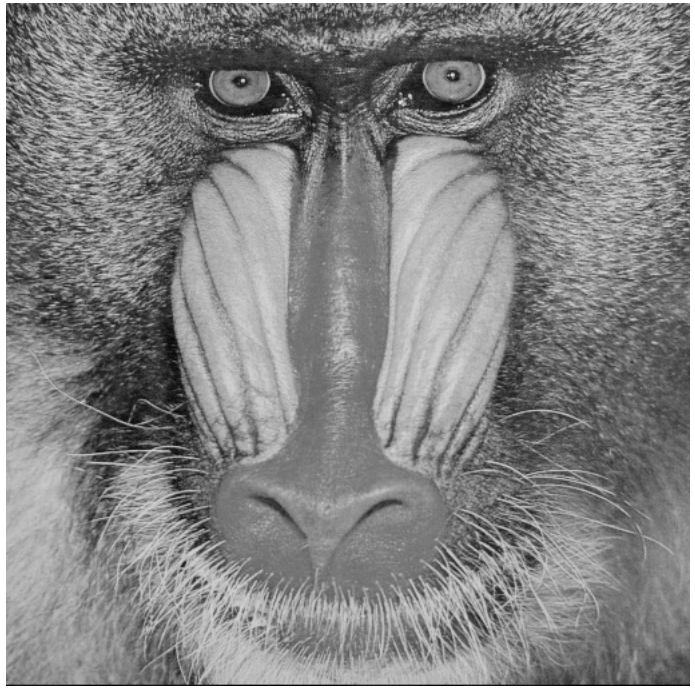
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If color is not required: In a first step: Color → Gray image

- Even better: if camera can provide the gray image directly
- less data to process
- fewer problems with many algorithms (e.g. median)



A standard test image: Mandrill (aka Baboon)
https://sipi.usc.edu/database/



color has vector representation
gray value is scalar

going from scalar to vector is often not possible

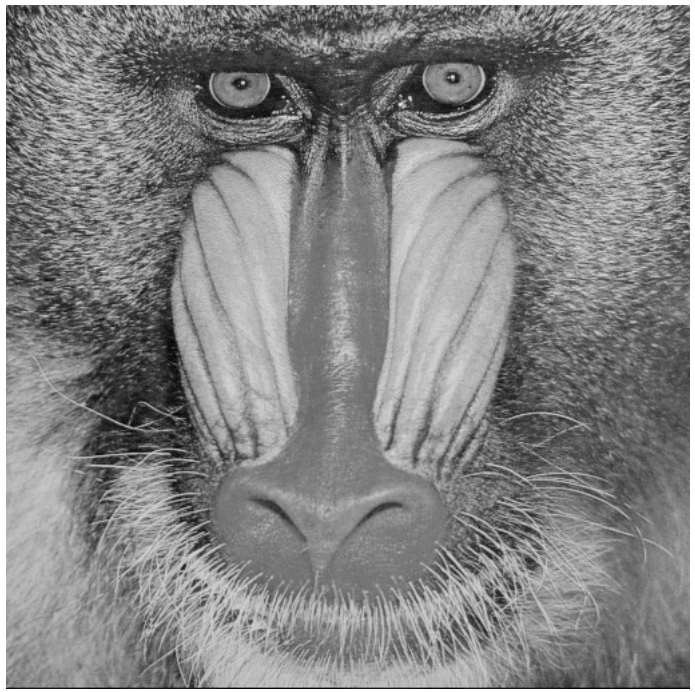
median: order value range and take the middle

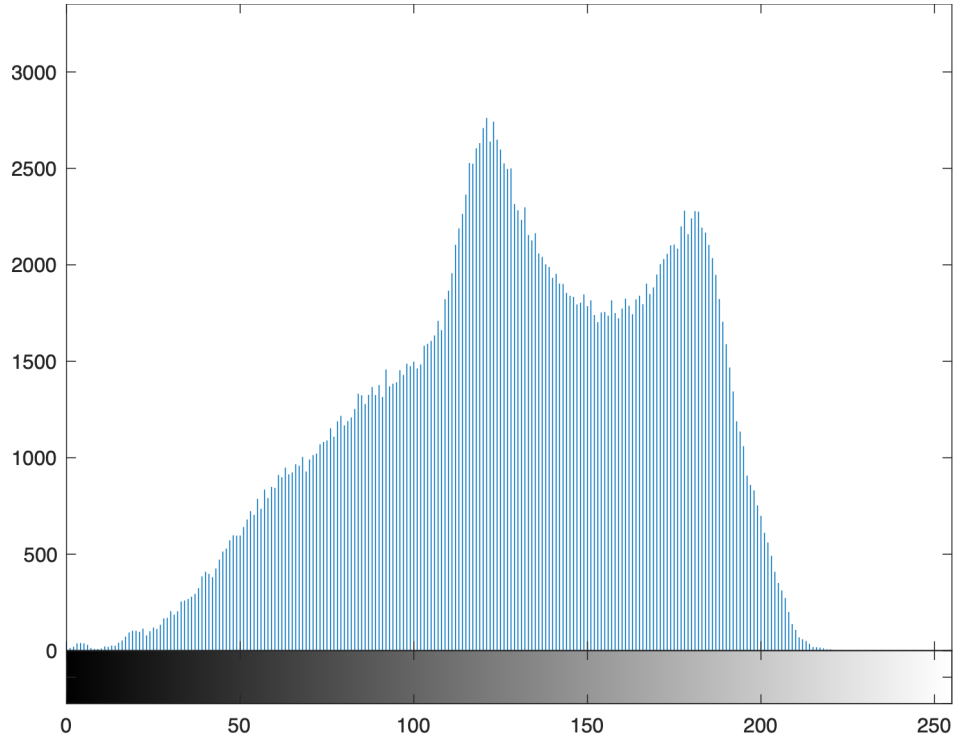
for color vector, not possible to sort them what determines if one vector is brighter than another?

Histograms and Thresholds

Histogram

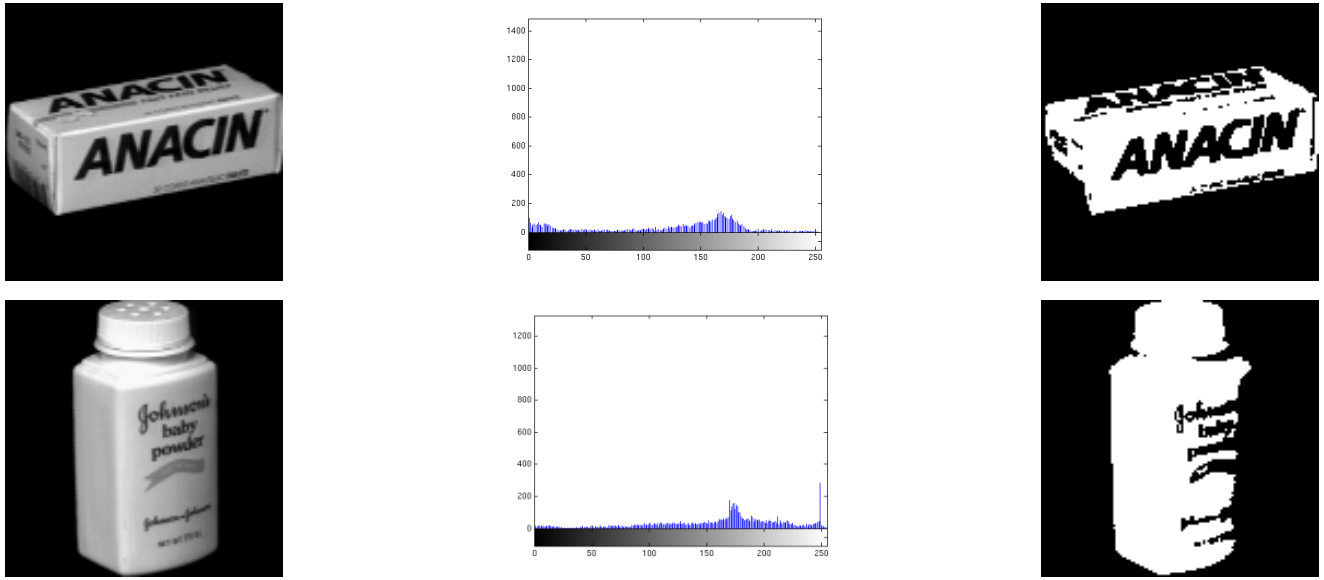
- shows the distribution of the gray values of an image
- horizontal axis: gray value
- Vertical axis: Number of pixels with this gray value





Binarization

- to separate (bright) objects from (dark) backgrounds → bimodal histogram
- Select threshold value t
  - if gray value ≤ t → 0 (black)
  - if gray value > t → 1 (white; binary; 255 for 8-bit gray value images;)



Pictures: COIL-20

Binarization – Influence of Threshold





t = 25



t = 76



t = 127



t = 178

Threshold Calculation

- bimodal histogram
  - Modeling by mixing two normal distributions
  - Threshold value: Intersection

assumption: values in image along two overlapping normal distributions and the valley between the distributions is the best target threshold

alpha is between 0 and 1, determines (in percentage) overlap between the two normal distributions

$$p(f) = \frac{\alpha}{\sigma_1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{f-\mu_1}{\sigma_1}\right)^2} + \frac{1-\alpha}{\sigma_2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{f-\mu_2}{\sigma_2}\right)^2}$$

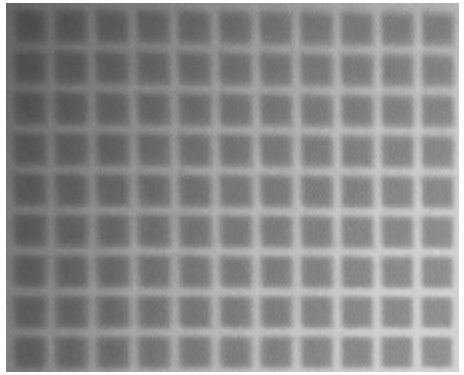
- Exact calculation of the parameters of the normal distributions and the intersection complex → approximate solution

Threshold Calculation – Algorithm

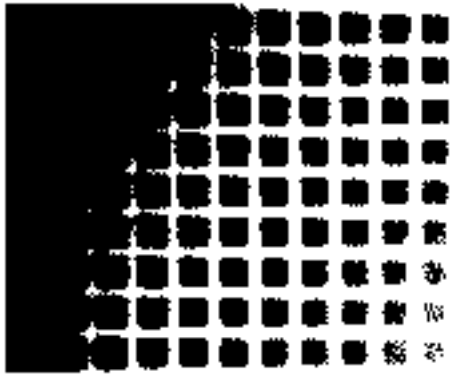
- Select start threshold t_0
  - Any value between minimum and maximum gray value
  - or mean value
  - or median
  - …
- Calculate parameters of normal distributions
  - σ_1, μ_1 from the values to the left of the threshold value
  - σ_2, μ_2 from the values to the right of the threshold value
  - α from ratio left/right part
- Calculate the residual error between histogram and mixture of normal distributions
- Change threshold value until error is minimal
- Gray values discrete → relatively few threshold values possible

$$p(f) = \frac{\alpha}{\sigma_1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{f-\mu_1}{\sigma_1}\right)^2} + \frac{1-\alpha}{\sigma_2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{f-\mu_2}{\sigma_2}\right)^2}$$

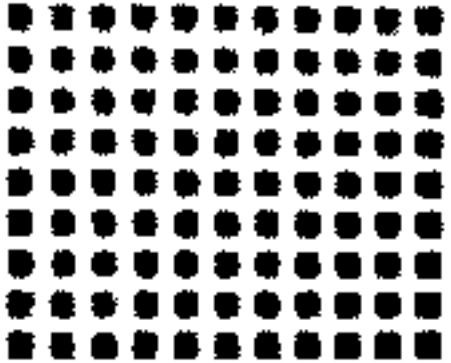
Adaptive Thresholds



Original image, inhomogeneously illuminated



Binarization using global histogram



Binarization using local histograms
(window of imagesize/8 x imagesize/8

Histogram Equalization

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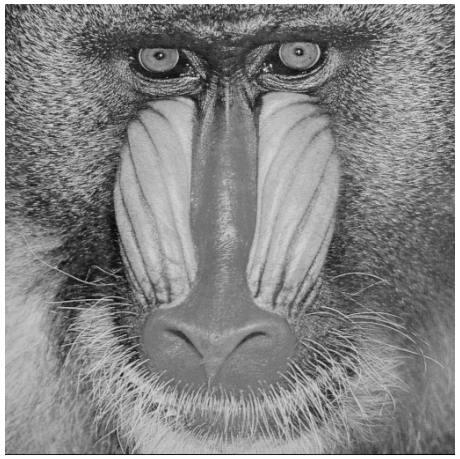
- Redistribution of the gray values so that the histogram is uniformly distributed
- Result:
  - Contrast enhancement
  - Reduced number of grayscales
- Given:
  - Histogram $H(i)$ of the original image (i = 0 ... 255)
  - Total number of pixels in the image M
- The new gray value g is calculated from the current one f :

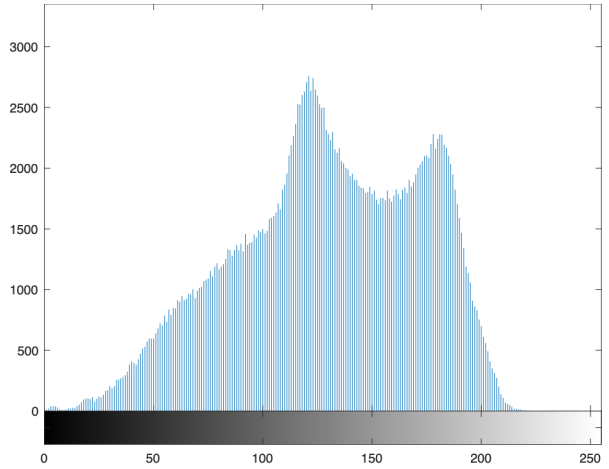
$$g = \frac{255}{M} \sum_{i=0}^{f} H(i)$$

Histogram Equalization Examples

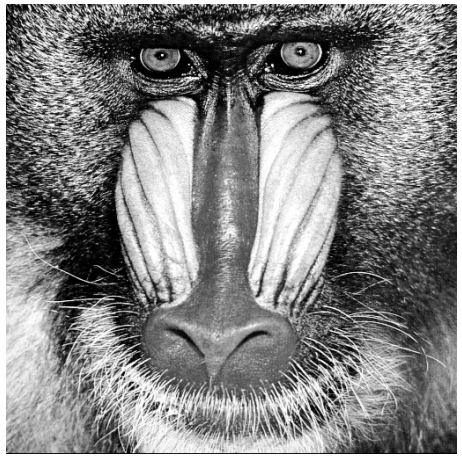
original image

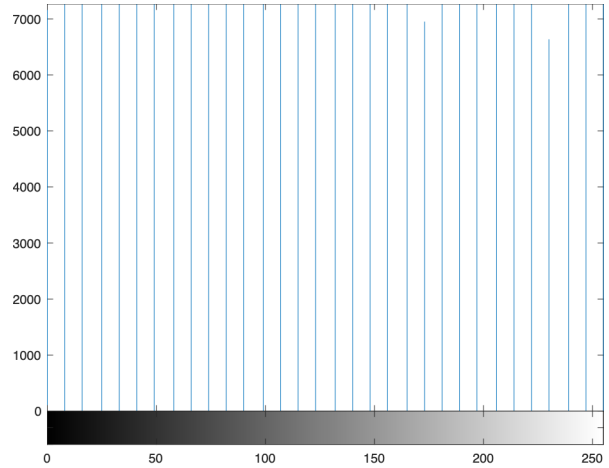
256 gray values



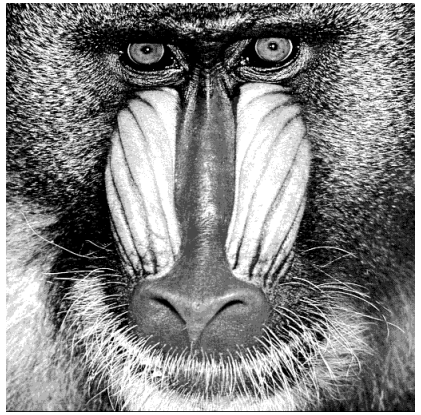


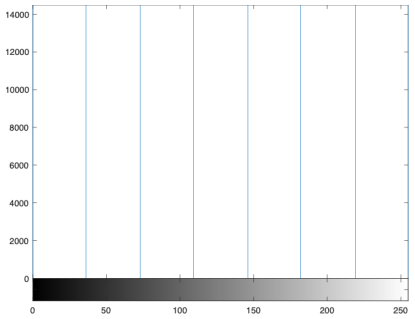
32 gray values



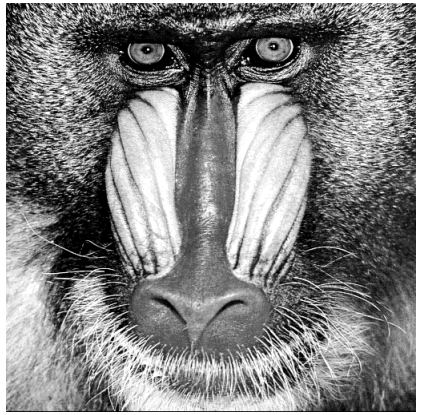


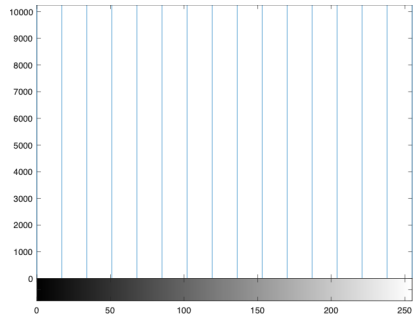
Histogram Equalization Examples



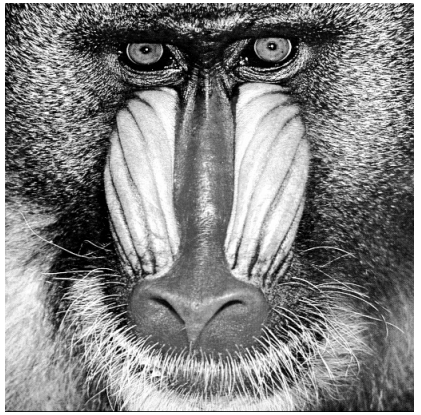


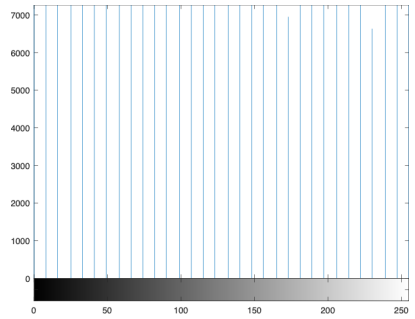
8 shades of gray



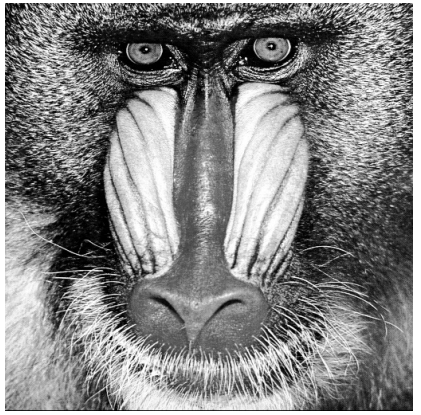


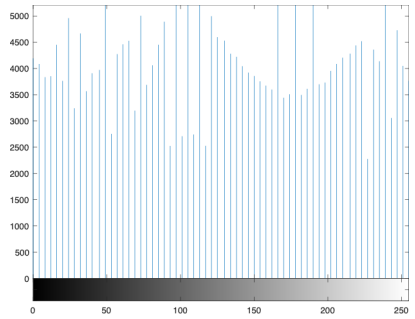
16 shades of gray



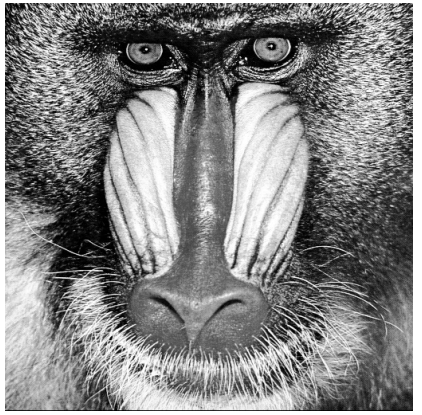


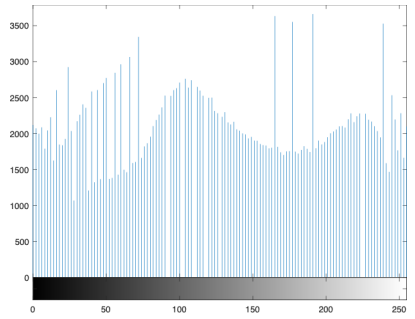
32 shades of gray





64 shades of gray

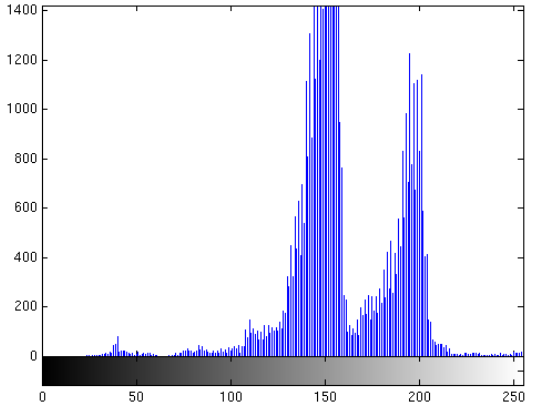




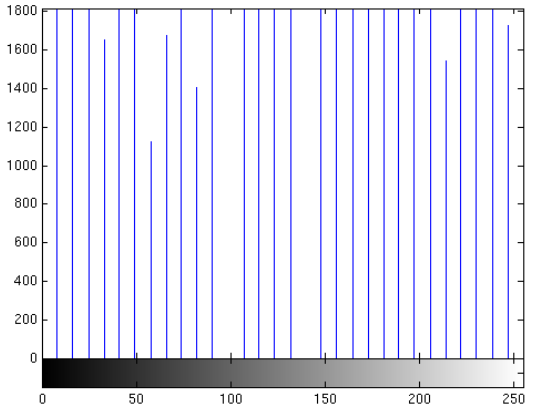
128 shades of gray

Histogram Equalization Examples







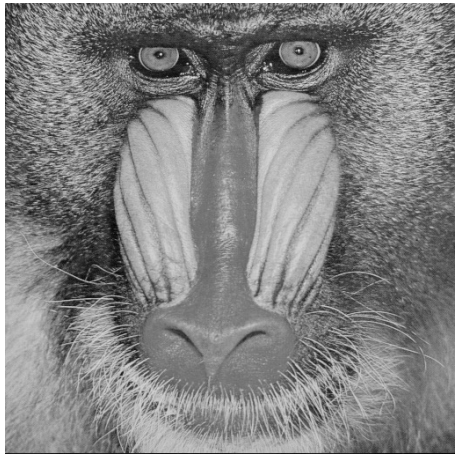


32 shades of gray

Linear Filters

Principle: Convolution (*Faltung*)

Original image f

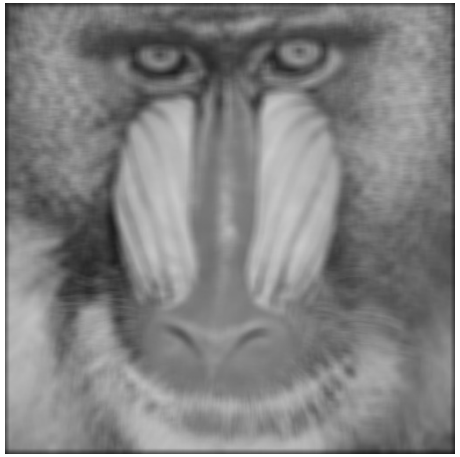


+ filter mask g

$$\begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix}$$

Convolution *

filtered image f'



$$f'(x,y) = f * g = \sum_{i=-r}^{r} \sum_{j=-r}^{r} f(x-i, y-j)\, g(i,j)$$

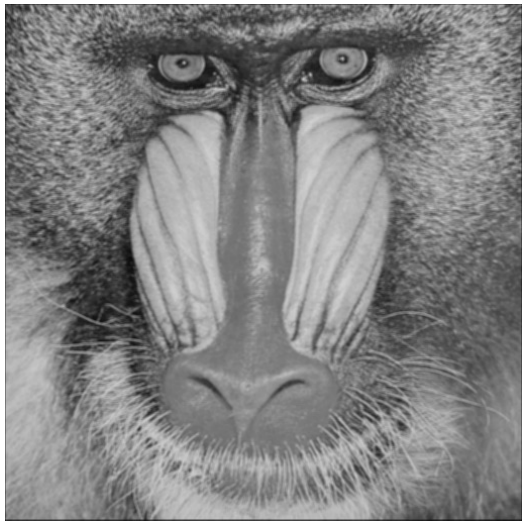
- Mirror filter mask horizontally and vertically,
- move filter mask over the image,
- calculate weighted sum mask/underlying gray value ⟶ new value for center pixel.

Filter mask g

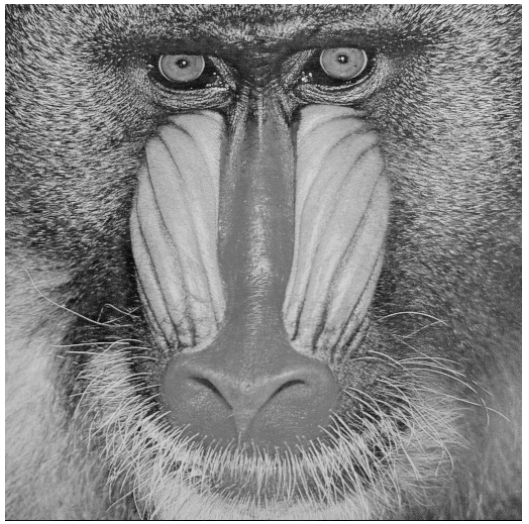
- Size $(2r+1) \times (2r+1)$
- (i,j)-coordinate system
- Origin (0, 0) in the center of the mask
- i to the right, j downwards (as usual in an image)

Low-pass Filter

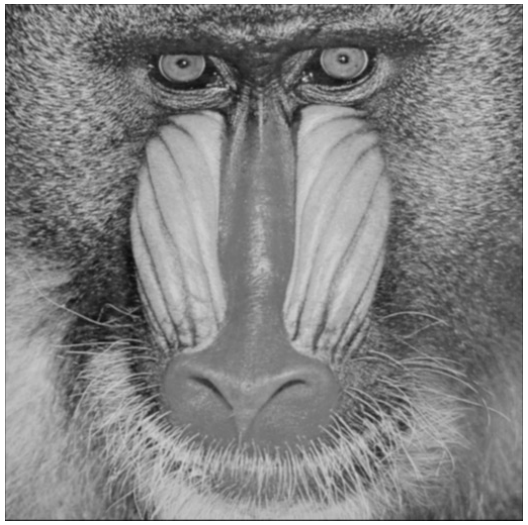
- Removes high frequencies
- Reduces noise
- Blurs the image



3x3 Mean $\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$

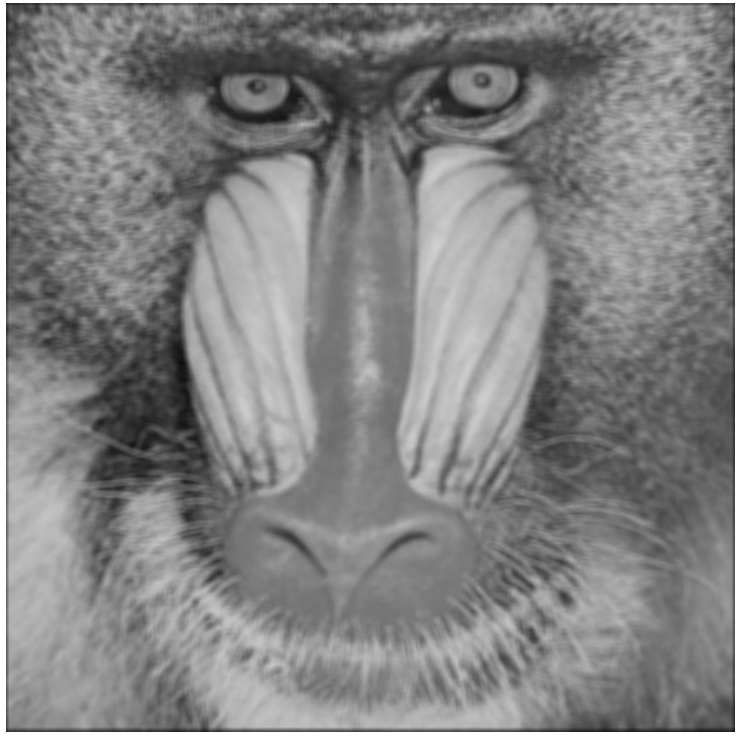


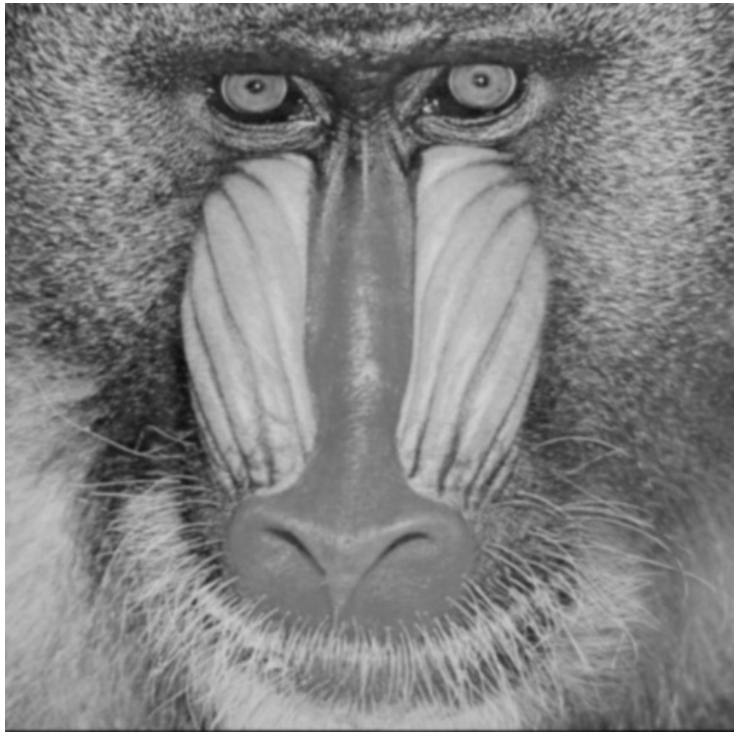
original image



3x3 Gauss $\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$

Low-pass filter

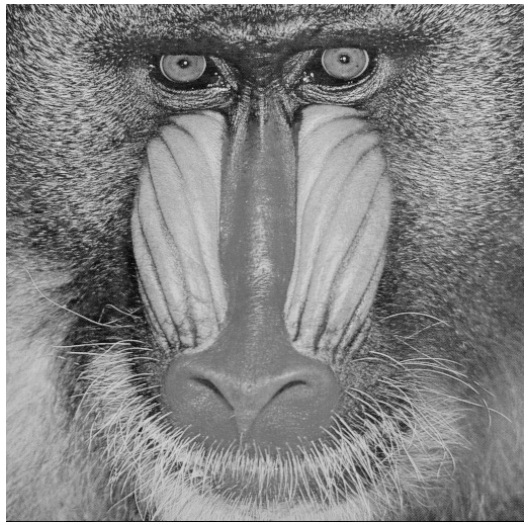




5x5 Mean $\frac{1}{25}\begin{bmatrix}1 & 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1 & 1\\ 1 & 1 & 1 & 1 & 1\end{bmatrix}$

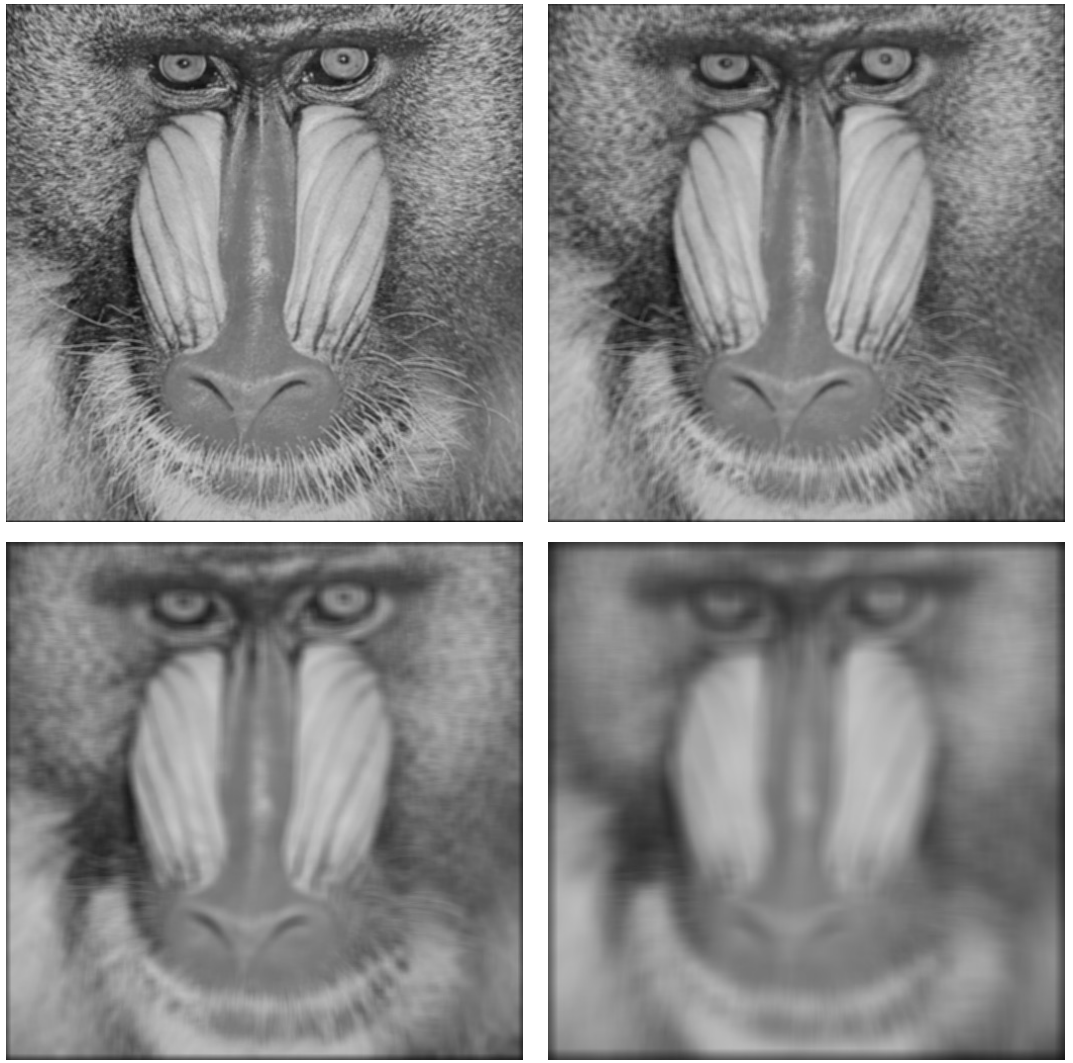
5x5 Gauss (with $\sigma = 1$) $\frac{1}{273}\begin{bmatrix}1 & 4 & 7 & 4 & 1\\ 4 & 16 & 26 & 16 & 4\\ 7 & 26 & 41 & 26 & 7\\ 4 & 16 & 26 & 16 & 4\\ 1 & 4 & 7 & 4 & 1\end{bmatrix}$

Influence of Filter Size



Original image

Mean filter, sizes: 3x3, 5x5, 11x11, 21x21



Color Images: Caution

- obvious method:
  filter each RGB color channel separately
  - can be done
  - however: RGB is unsuitable for linear interpolations
- better: use CIELUV or CIELAB
  - but: Conversion computationally very expensive (non-linear)
- CNNs will apply a 3D-filter mask (tensor)
  - this results in a single channel as output



Mean 11x11 on RGB



Gauss 11x11, $\sigma = 5$ on RGB

Edge Filters

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- High-pass (removes low frequencies) combined with low-pass (smoothing)
- Edge detection
- Widespread:
  - Sobel
    - based on calculation of the first partial derivatives
    - results in two edge images (horizontal and vertical direction)
    - Edges = large values (maxima of the first derivative)
  - Laplace
    - based on calculation of the second derivative
    - Edges = zero crossings
    - More susceptible to noise than Sobel

Sobel

- first partial derivatives

$$f_x = \frac{\partial f(x,y)}{\partial x} \qquad \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} \qquad f_y = \frac{\partial f(x,y)}{\partial y} \qquad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

- Gradient strength:

$$s = \sqrt{f_x^2 + f_y^2}$$

Value range?

- Gradient direction:

$$\theta = \arctan\left(\frac{f_y}{f_x}\right)$$

Note: use atan2(y, x)

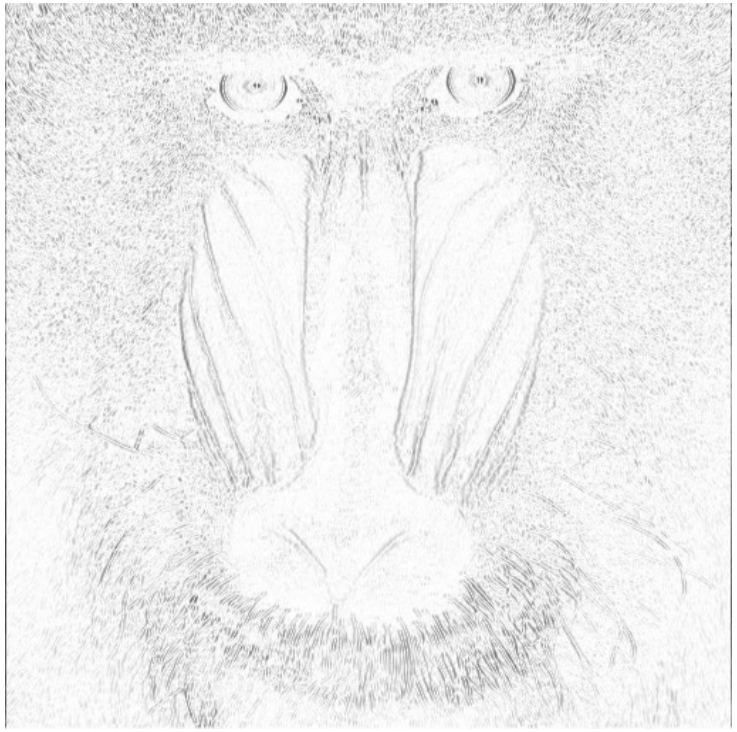
Edge Filter – Sobel

horizontal



$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

vertical



$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

(images scaled & inverted)

Laplace

- Laplace operator from 2nd order partial derivatives

$$\Delta f(x,y) = \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}$$

- different variants

8-Neighborhood

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

4-Neighborhood

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

with smoothing

$$\begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Value range?

Edge Filter – Laplace



Laplace $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

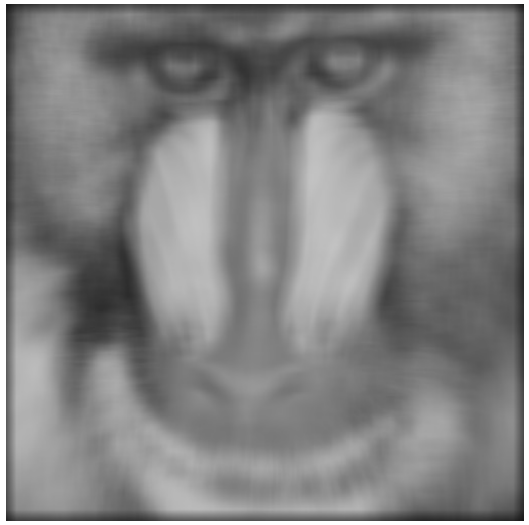


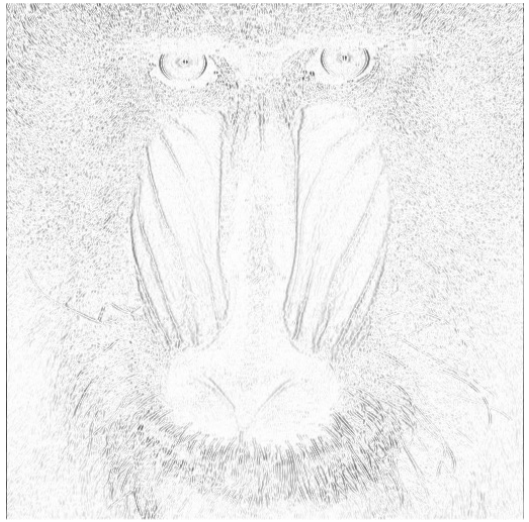
combined Sobel
(gradient magnitude image)

(images scaled & inverted)

Padding

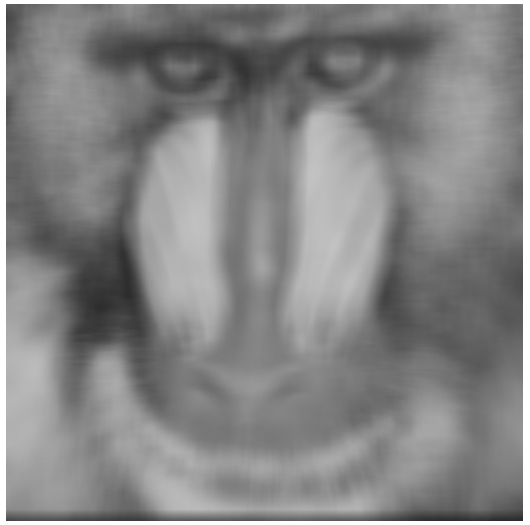
Don’t want these black borders?

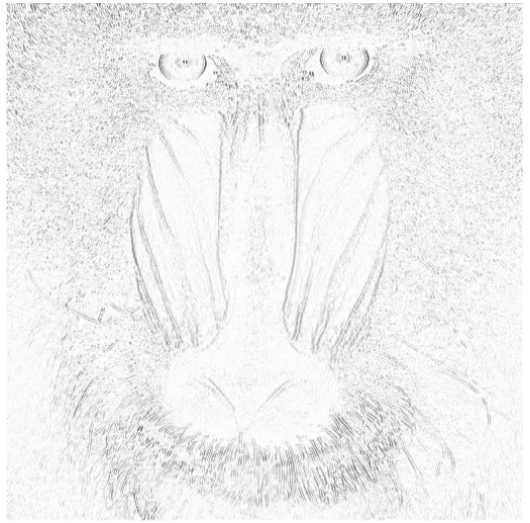




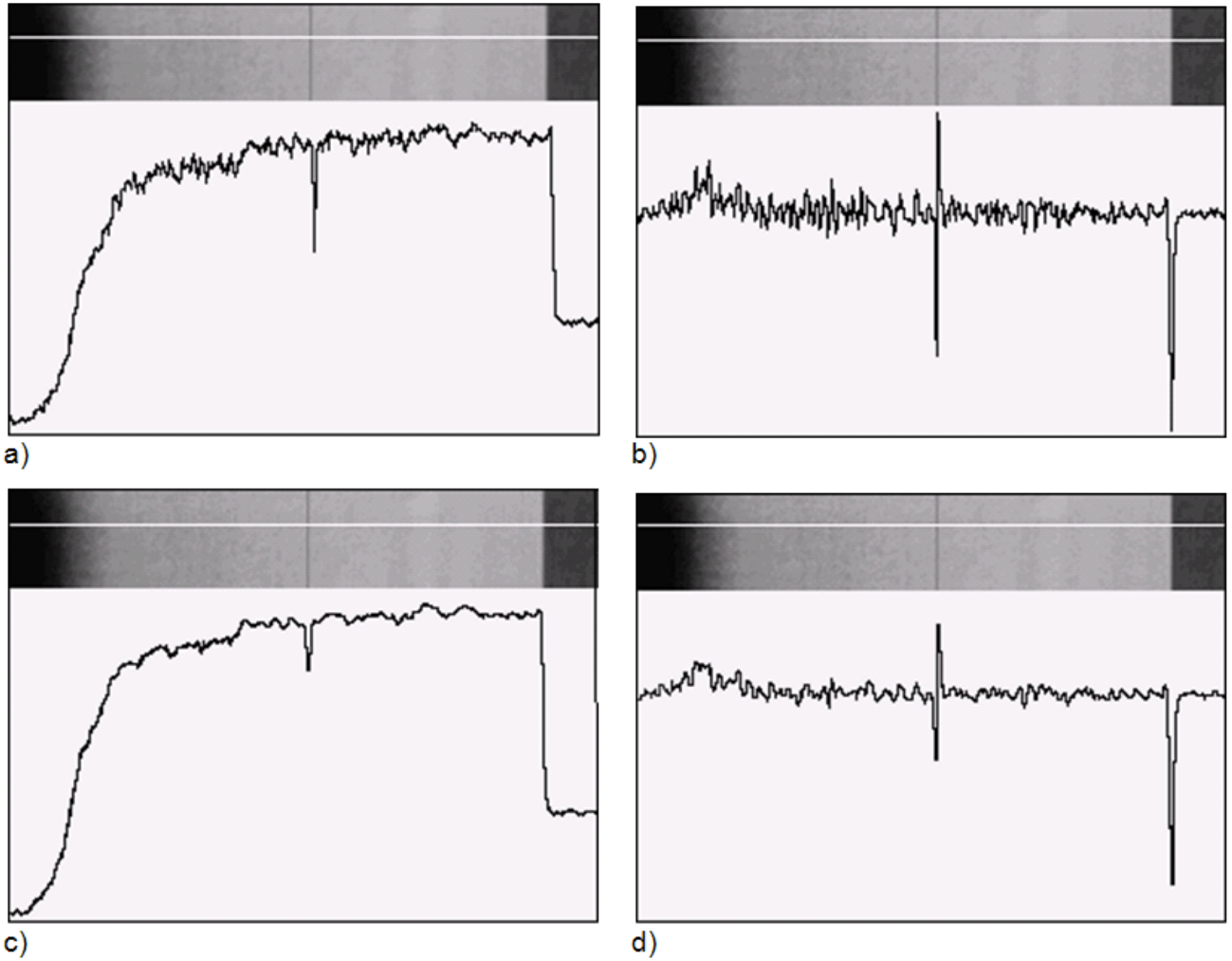
Options:

- decrease image dimensions
- use special filter masks at border
- padding
  - replicated
  - mirrored/reflected
  - periodic





Filter – Section through Image



from: H. Ernst, Industrial applications of image processing

a) Original image
b) Gradient filter
c) Low pass
d) Low pass & Gradient filter

Notes

- Filter masks typically have odd sizes (3x3, 5x5, ...) → symmetrical around current pixel
- they are not necessarily square
- there are other widely used filter masks that have not been presented here (Prewitt, Harris, Robert's Cross, ...)

- Design of filters for special applications possible
- Filters can be cascaded by combining them in **a single** mask (see convolution formula)

- Convolution is normally carried out in the spatial domain
- Convolution with FFT in frequency domain only makes sense with large masks

Non-linear Filters

Morphological Operations

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- We now consider binary images (0 and 1)
- Morphological operation = operation on sets
  - Image
  - Structure element (defines the neighborhood)
- basic operations
  - Erosion
  - Dilation
- and the combinations thereof
  - Opening
  - Closing

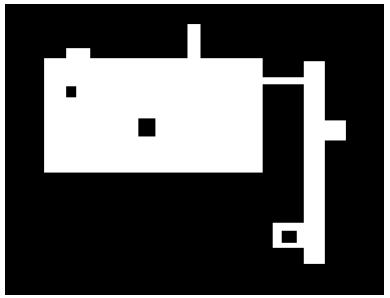
Erosion

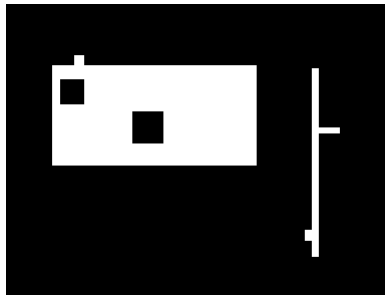
- Erosion of the binary image $\boldsymbol{f}$ with structure element $\boldsymbol{s}$
- $\boldsymbol{h = f \ominus s}$

$$h_{ij} = \begin{cases} 1 & \text{wenn } f_{i+\mu,j+\nu} = 1 \quad \forall(\mu,\nu) \in \boldsymbol{s} \\ 0 & \text{sonst} \end{cases}$$

- resp.

$$h_{ij} = \min_{(\mu,\nu)\in \boldsymbol{s}} \{f_{i+\mu,j+\nu}\}$$





Dilation

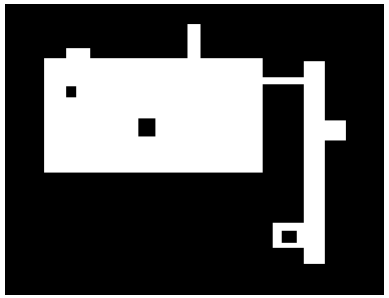
- Dilation of the binary image $\boldsymbol{f}$ with structure element $\boldsymbol{s}$

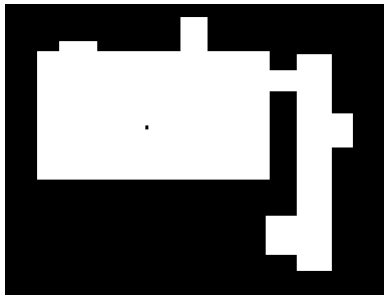
$$\boldsymbol{h} = \boldsymbol{f} \oplus \boldsymbol{s}$$

$$h_{ij} = \begin{cases} 1 & \text{wenn} \quad \exists(\mu,\nu) \in \boldsymbol{s} \text{ mit } f_{i+\mu,j+\nu} = 1 \\ 0 & \text{sonst} \end{cases}$$

- resp.

$$h_{ij} = \max_{(\mu,\nu)\in \boldsymbol{s}} \{f_{i+\mu,j+\nu}\}$$

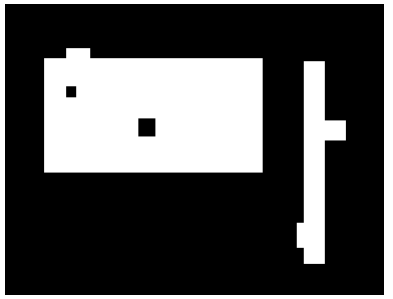


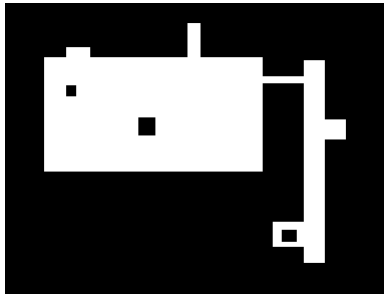


Opening and Closing

- The morphological operations Opening and Closing result from cascading erosion and dilation
- Opening

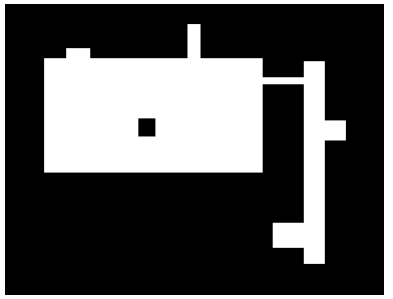
$$\boldsymbol{h} = (\boldsymbol{f} \ominus \boldsymbol{s}) \oplus \boldsymbol{s}$$





- Closing

$$\boldsymbol{h} = (\boldsymbol{f} \oplus \boldsymbol{s}) \ominus \boldsymbol{s}$$



Rank Order Operations

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- now: back to grayscale images
- **sort** gray values within the neighborhood defined by the structure element

- This results in a ranking of the gray values

$$R_{ij} = \{r_1, r_2, \ldots, r_M \mid r_k \leq r_{k+1} \forall k = 1, \ldots, M\}$$

- Rank order operation: Function on R_{ij}

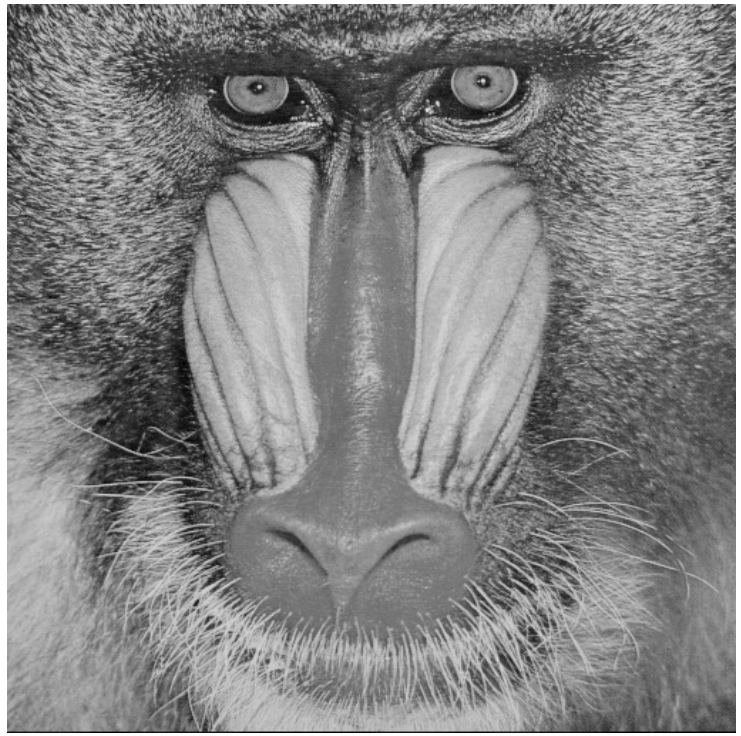
$$h_{ij} = g(R_{ij})$$

Rank Order Operations – Examples

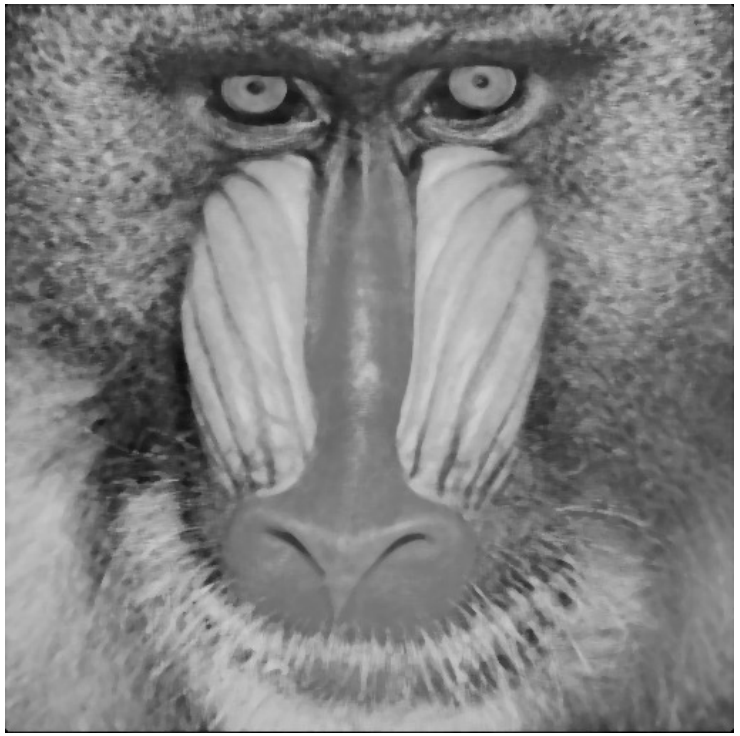
- Median $h_{ij} = r_{(M+1)/2}$
- Erosion $h_{ij} = r_1$
- Dilation $h_{ij} = r_M$
- Contour extraction $h_{ij} = r_M - r_1$
- Contrast enhancement

$$h_{ij} = \begin{cases} r_1 & \text{wenn} \quad f_{ij} - r_1 < r_M - f_{ij} \\ r_M & \text{sonst} \end{cases}$$

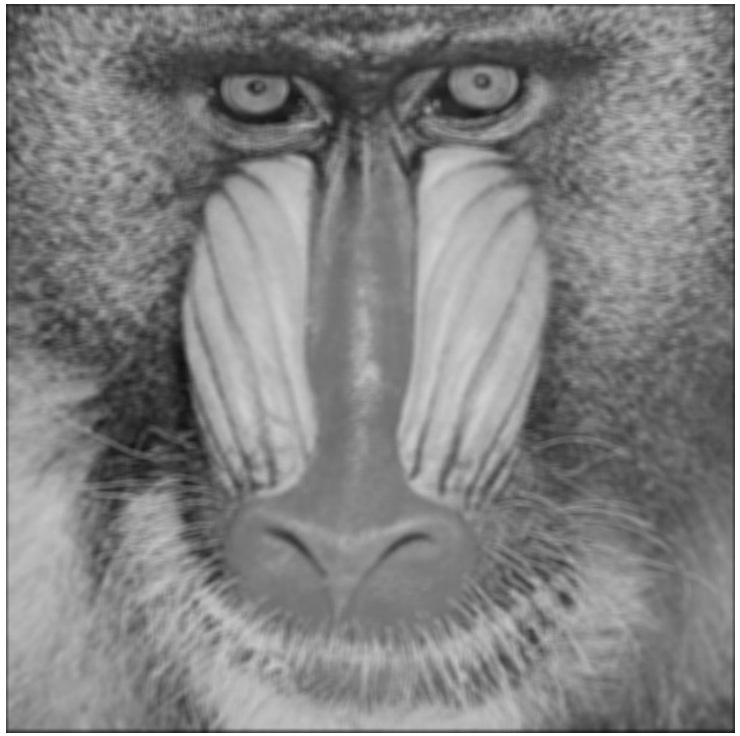
Comparison Median – Low-pass Filter



Original



5x5 Median



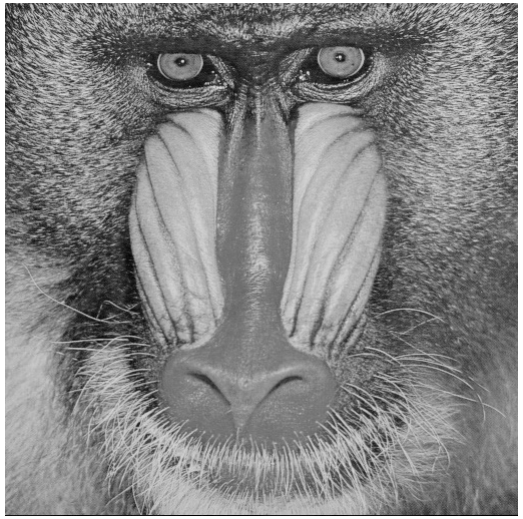
5x5 Mean

Median is edge-preserving!

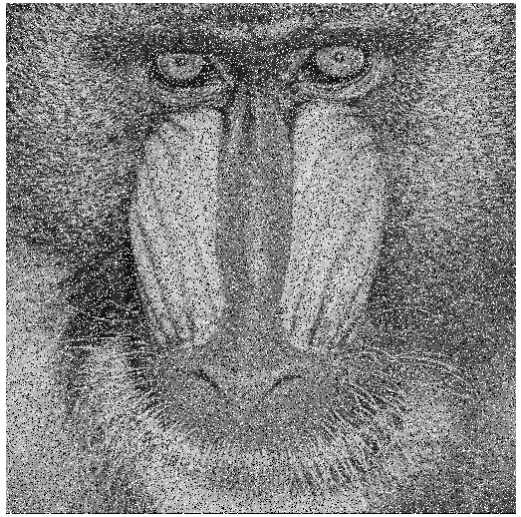
Comparison Median – Low-pass Filter

Technische Hochschule Rosenheim

Original



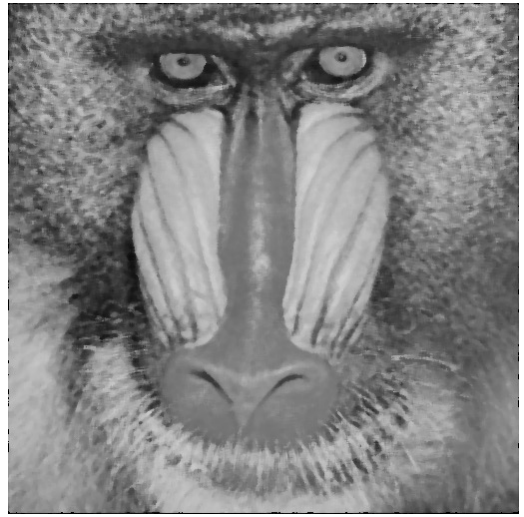
Noisy



5x5 Mean



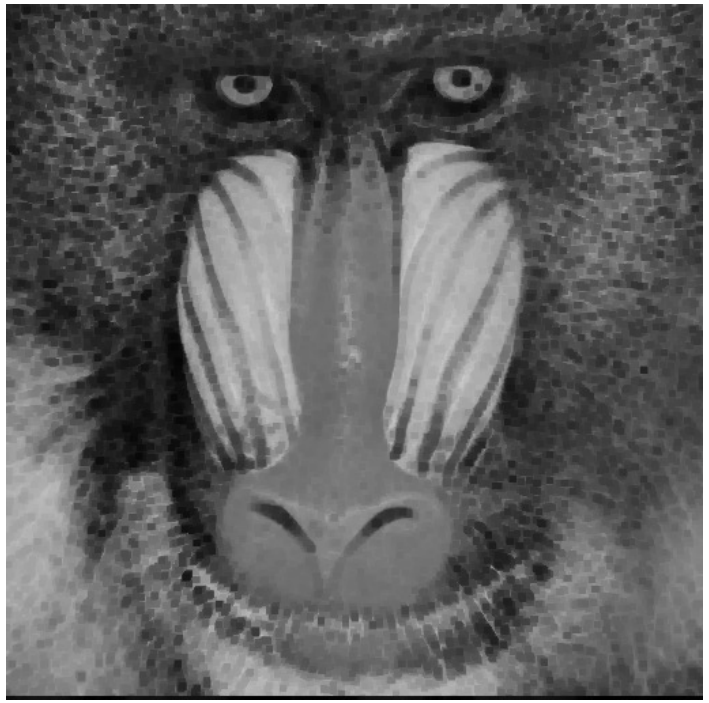
5x5 Median



Grayscale Erosion/Dilation/Contour (5x5)

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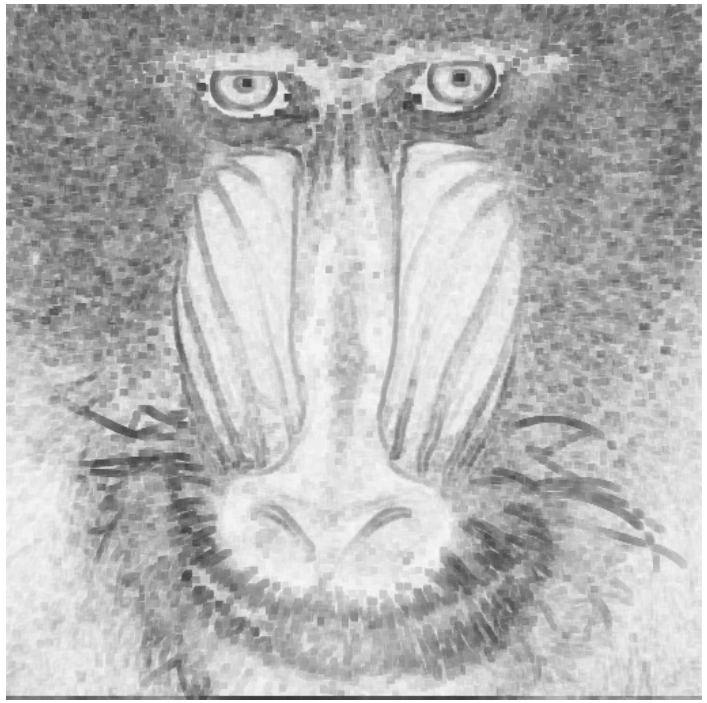
$h_{ij} = r_1$



$h_{ij} = r_M$



$h_{ij} = r_M - r_1$ (inverted)



Color Images?

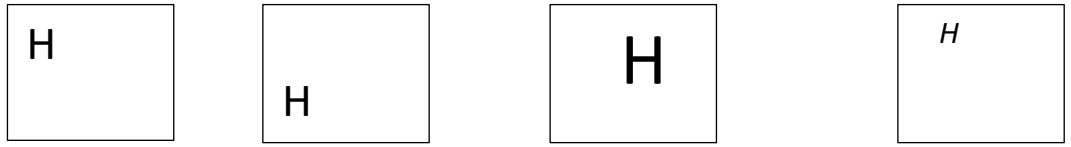
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- Low-pass filter
  - often: separate processing of the color channels
  - Color space: better not RGB but, e.g., CIELAB, CIELUV
- High-pass filter
  - Separate processing of the color channels makes no sense for edge detection
- linear (high & low pass) filters are used in CNN:
  - the values of the filter masks are learned from training data
  - instead of 2D mask: 3D tensor
  - this means that all color channels are processed together
  - mostly: RGB; other color spaces would also be more suitable here
- Rank order filter/median: difficult, as
  - color channels should not be processed separately (this is no longer a median)
  - vectors have no order (but there are approaches for this)

Normalization

Objective

- Patterns can vary in many parameters that are irrelevant for class membership, e.g.



- Objective: Mapping of these parameters to standard values (facilitation/improvement of subsequent processing)
- Examples:
  - Size
  - Rotation
  - Position
  - Brightness/Lighting
  - Color
  - ...

Attention!

Technische Hochschule Rosenheim

- Only **irrelevant** parameters may be normalized

- Example 1
  - The size of H and $_{H}$ is not relevant
  - The size of / and ´ is relevant

- Example 2
  - The rotation of a red blood cell is not relevant
  - The rotation of 6 and 9 or M and W is relevant

Size

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Size normalization is achieved by **scaling** the image to a standard size (resampling)

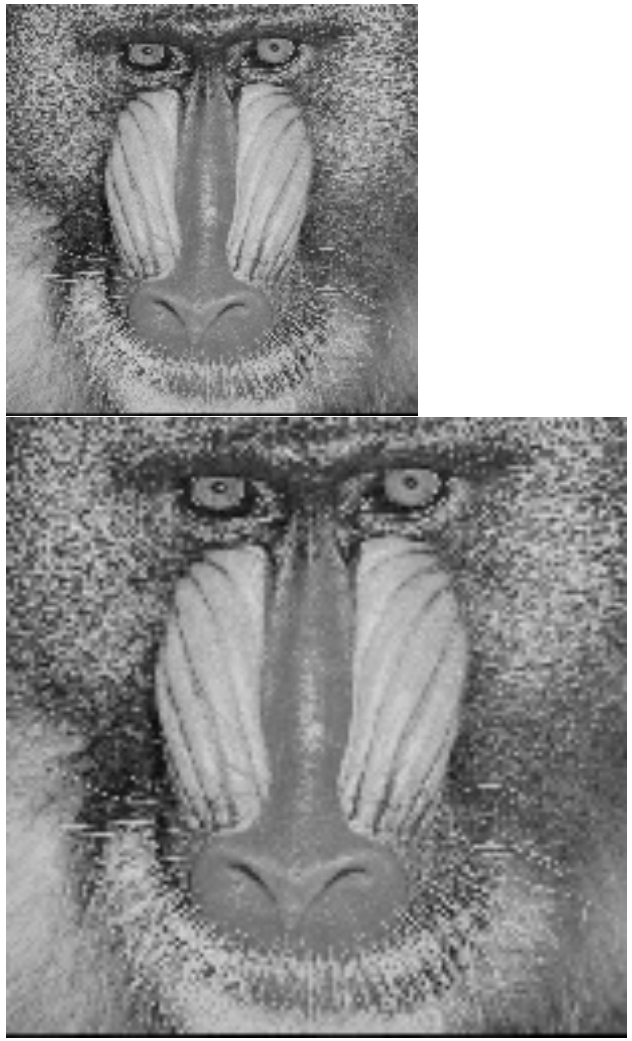
- optimal: Use the reconstruction formula of the sampling theorem
  - Disadvantage: computational effort, as sinc only decays slowly
- Nearest neighbor interpolation
  - very fast
  - leads to block structures
- Bilinear interpolation
  - blurs contrasts
- B-Splines
  - 1st degree = bilinear interpolation
  - 3rd degree (cubic B-splines) often used for high quality results

- Nearest neighbor/bilinear interpolation is often sufficient for classification

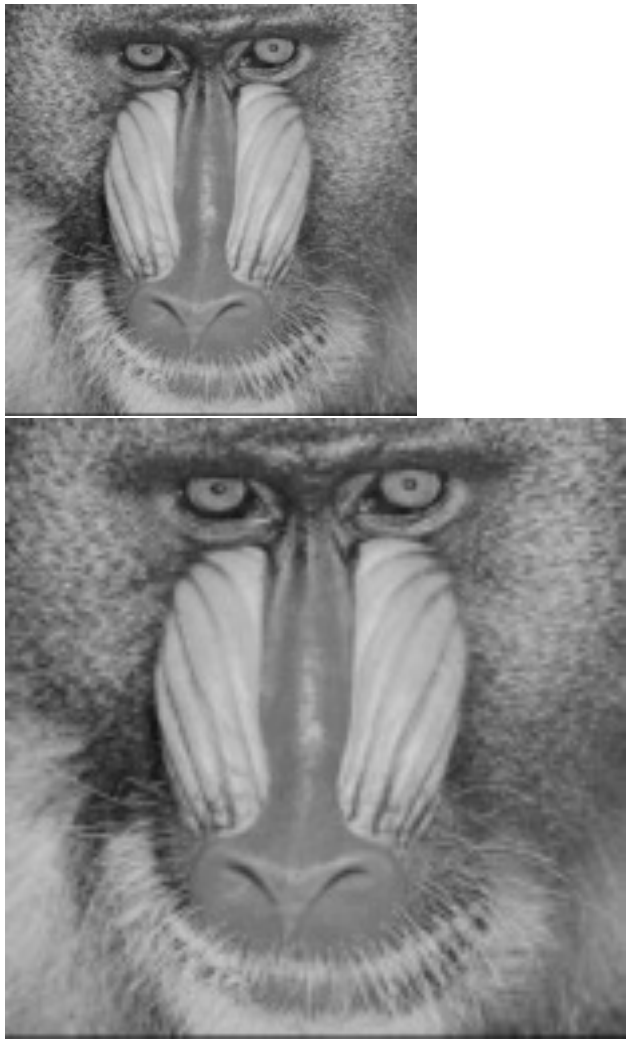
Scaling – Example

Technische Hochschule Rosenheim

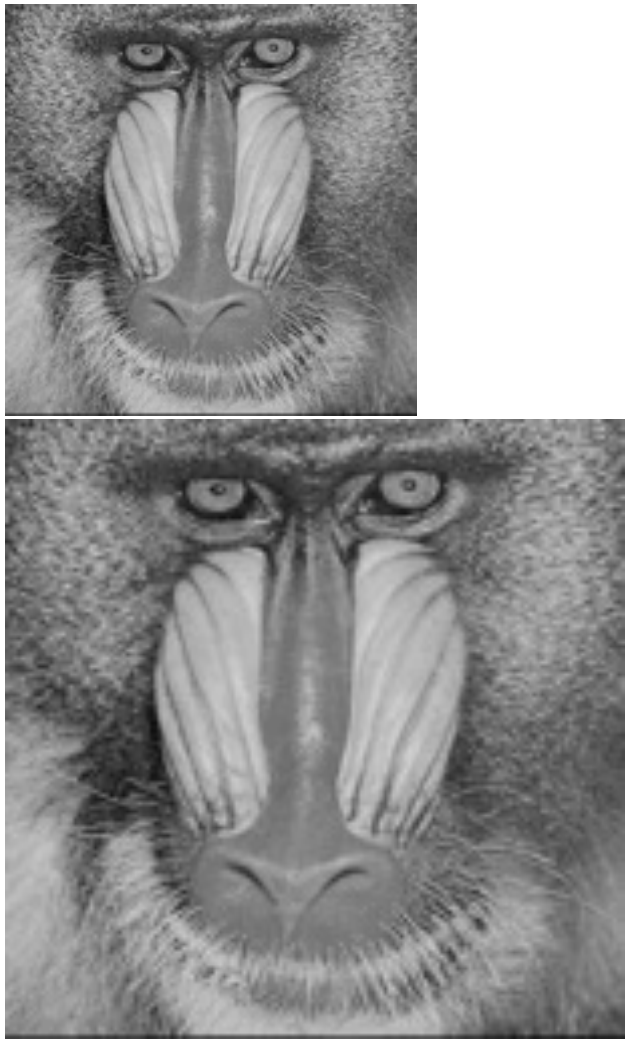
Nearest neighbor



Bilinear



Bicubic

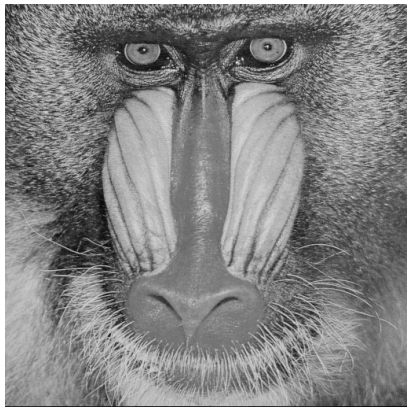


Brightness

Technische Hochschule Rosenheim

Can be achieved by

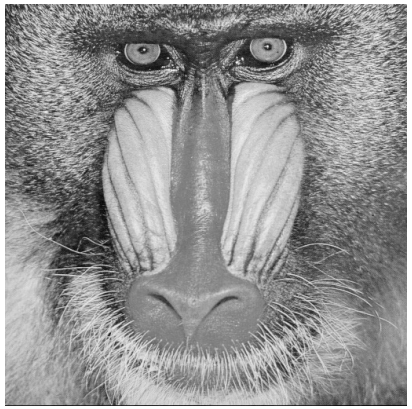
- Stretching the gray values
- Normalization of mean value and variance of grey values



min = 0
max = 231

Stretching the gray values to interval $0 \leq f_j{}' \leq \alpha$

- with $\alpha = 1$ or $\alpha = 2^N - 1$ (N number of bits for quantization)
- $f_{\min}$ / $f_{\max}$ smallest or largest gray value occurring in the image



min = 0
max = 255

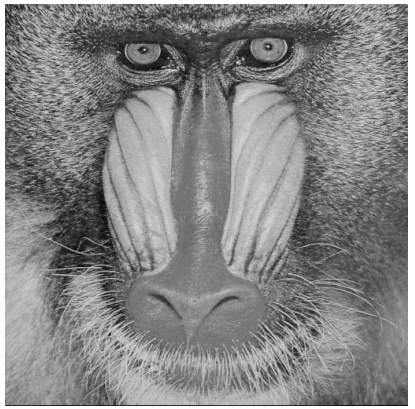
$$f_j' = \alpha \frac{f_j - f_{\min}}{f_{\max} - f_{\min}}$$

Brightness – Mean and Variance

- Normalization of the grey values to mean 0 and variance 1

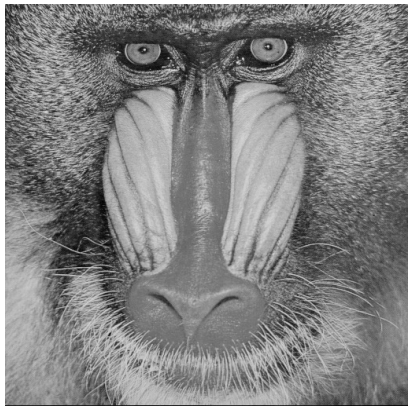
$$\mu = \frac{1}{M}\sum_{j=0}^{M-1} f_j \qquad \sigma^2 = \frac{1}{M}\sum_{j=0}^{M-1} f_j^2 - \mu^2$$

$$f_j' = \frac{f_j - \mu}{\sigma}$$



$\mu = 129.6$

$\sigma = 42.3$



$\mu = 128$

$\sigma = 42.6$

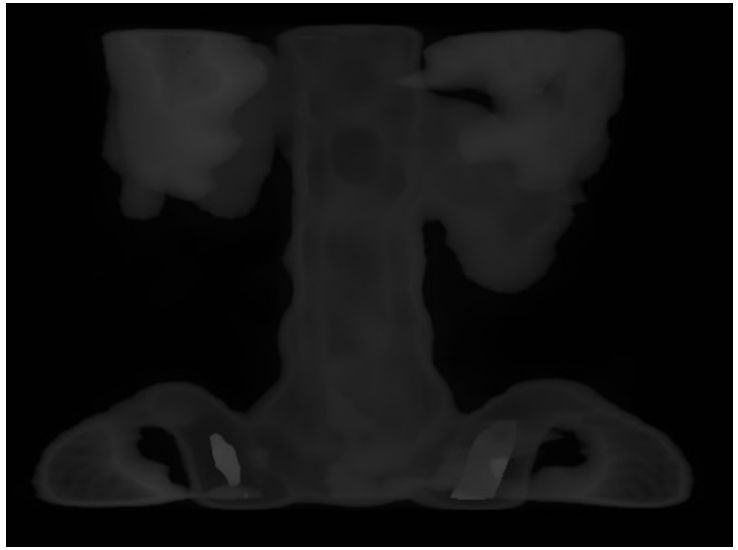
- invariant against affine linear transformations $af_j + b$, a $\neq 0$
- often, with N bits per gray value
  - Mean value $b = 2^{N-1}$ (8 bit: 128)
  - standard deviation $a = 2^{N-1} / 3$ (3σ interval)
    limit values that are too large/small

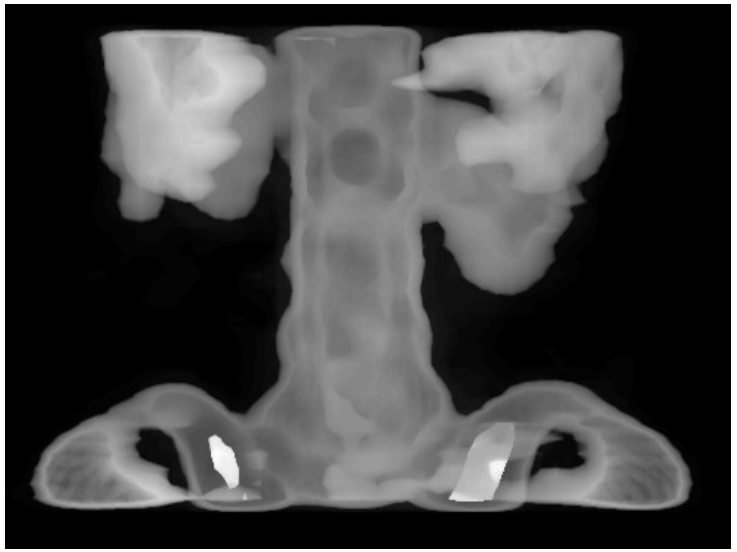
Brightness – Example

min = 1
max = 64

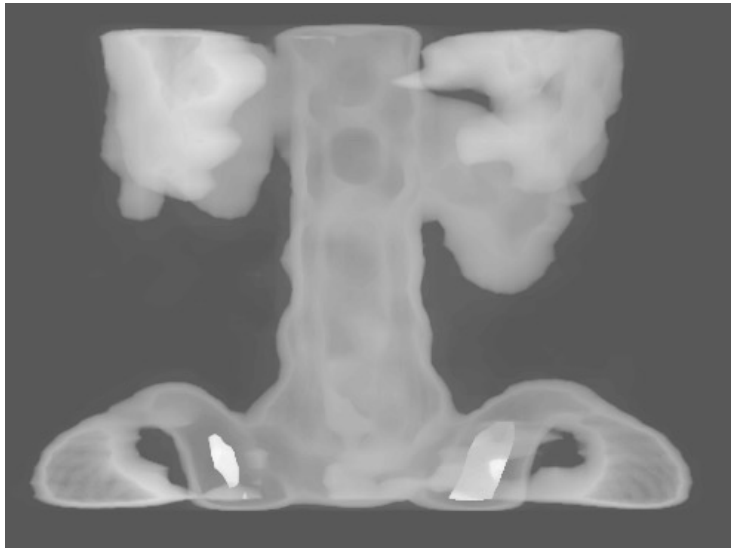
$\mu = 16.7$

$\sigma = 16.6$





min = 0
max = 255

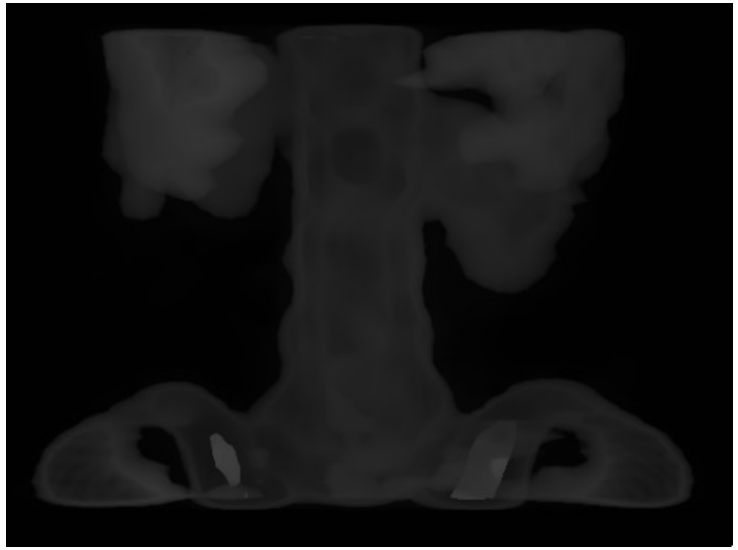


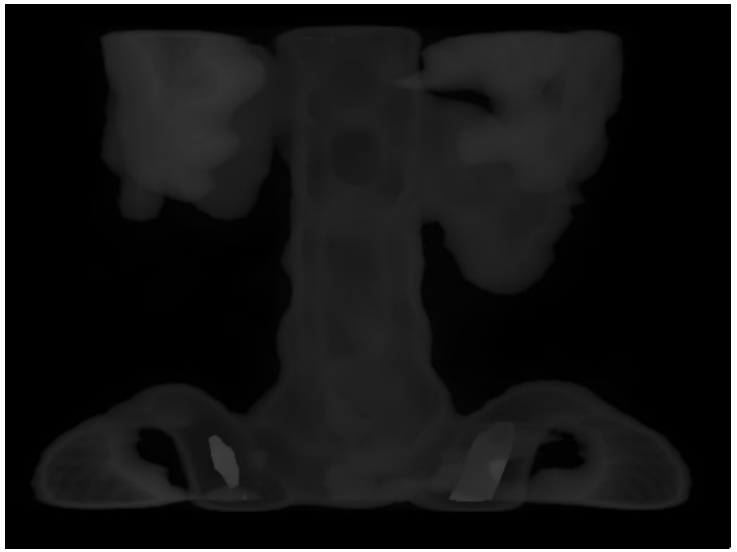
$\mu = 128$

$\sigma = 42.7$

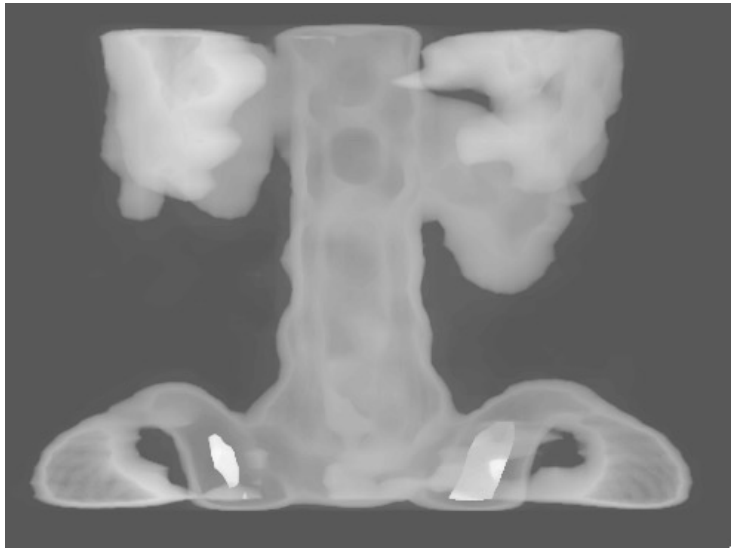
Brightness – Example 2

Oops – I modified only a single pixel in the original image…





min = 0
max = 255



$\mu = 128$

$\sigma = 42.6$

Moments

- from statistics
- (discrete) moment of order (p+q) in 2D (images):

$$m_{pq} = \sum_{x=0}^{M_x-1} \sum_{y=0}^{M_y-1} x^p y^q f(x,y)$$

- Center of gravity:

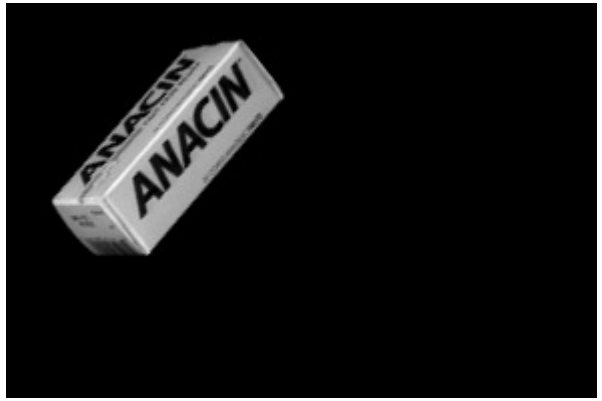
$$x_s = \frac{m_{10}}{m_{00}}, \qquad y_s = \frac{m_{01}}{m_{00}}$$

- Central moments (normalized to center of gravity):

$$\mu_{pq} = \sum_{x=0}^{M_x-1} \sum_{y=0}^{M_y-1} (x-x_s)^p (y-y_s)^q f(x,y)$$

Center of Gravity

Original



Center of gravity (on gray values)



(81.762; 73.799)



binarized



(81.429; 74.600)

Center of gravity (on binary image)

Moments (2)

- Normalization 1. central moments: Mean value (shift to the center of gravity)

$$x' = x - x_s, \quad y' = y - y_s, \quad \Rightarrow \quad \mu_{01} = \mu_{10} = 0$$

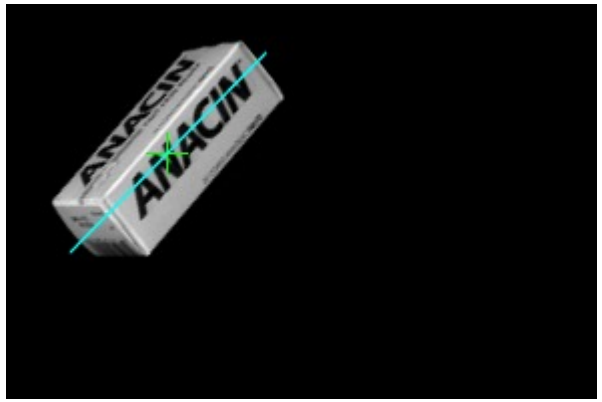
- Normalization 2. central moments: variance (“size”)

$$x' = \frac{x}{\sqrt{\mu_{02} + \mu_{20}}}, \quad y' = \frac{y}{\sqrt{\mu_{02} + \mu_{20}}}, \quad \Rightarrow \quad \mu_{02} + \mu_{20} = f(x, y)$$

- Normalization 2. central moments: Skew (rotation to principal axes)

$$\tan 2\alpha = \frac{2\mu_{11}}{\mu_{20} - \mu_{02}} \qquad \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \Rightarrow \quad \mu_{11} = 0$$

Rotation Angle



$\mu_{11} = -2054848.4$

$\mu_{20} = 2926618.9$

$\mu_{02} = 3007790.4$

calculated on binarized images

$$\tan 2\alpha = \frac{2\mu_{11}}{\mu_{20} - \mu_{02}} = \frac{-4109696.8}{-81171.5} = 50.63$$

$\alpha = -45.57°$



$\mu_{11} = -520193$

$\mu_{20} = 5050284$

$\mu_{02} = 1008464$

$$\tan 2\alpha = \frac{2\mu_{11}}{\mu_{20} - \mu_{02}} = \frac{-1040386}{4041820} = -0.2574$$

$\alpha = -7.22°$

Caution:

- For arctan: use atan2(y, x)
- y-axis points downwards!

Notes

- Normalization of higher order moments is possible
- Center of Gravity:
  - when using gray values: Center of mass, shifted towards light values
  - for geometric center of gravity: object pixel = 1, background = 0 (binarized)

Summary

Technische Hochschule Rosenheim

- Histograms
  - Threshold calculation for binarization of images
  - Histogram equalization (contrast enhancement)
- linear filters
  - Convolution, low-pass (mean value, Gaussian), edge filter (Sobel, Laplace)
- non-linear filters
  - Morphological operations
    (erosion, dilatation, opening, closing)
  - Rank order operations (median)
- Normalization measures
  - Size
  - Brightness
  - Moments (center of gravity, variance, rotation)